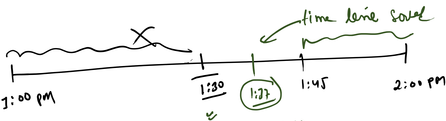
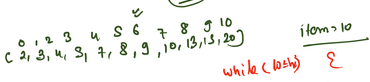
Lec-4
26/08/26 2:34 PM





item = 10

while (lo<hi) {

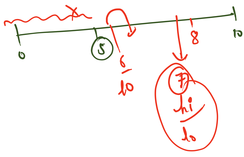
mid = (lo+hi) / 2

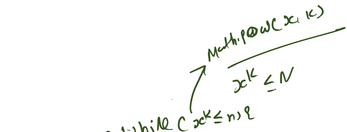
if (arr[mid] == item) {
item mid ✓

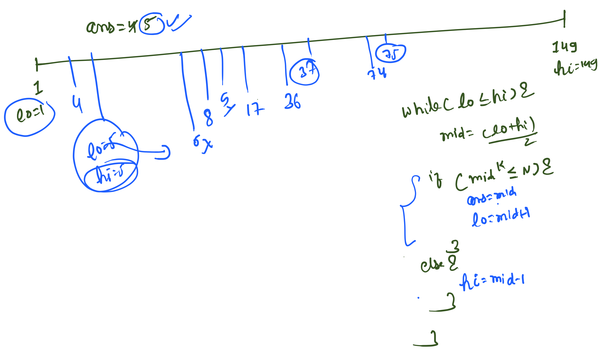
} else if (arr[mid] > item) {
hi = mid-1

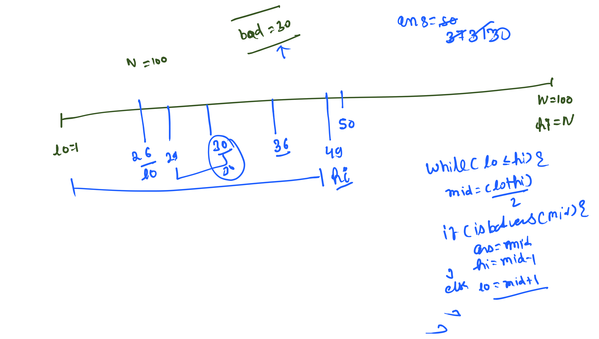
} else
lo = mid+1

}










```
public int firstBadVersion(int n) {
    int lo = 1;
    int hi = n;
    int ans = 0;
    while (lo <= hi) {
        int mid = (lo + hi) / 2;
        if (isBadVersion(mid)) {
            ans = mid;
            hi = mid - 1;
        } else {
            lo = mid + 1;
        }
    }
    return ans;
}
```

lo/2 + hi/2

N = 2^31 - 1 ✓

bad = 2^31 - 3 ✓

lo = 2^31 - 6

hi = 2^31 - 1

lo + hi/2 = (10 + (hi - lo)) / 2 = (2^31 - 6 + 2^31 - 1) / 2 = (2x2^31 - 7) / 2

